

5.18 解:

$${}^0_3\vec{O} = \begin{bmatrix} l_1 c_1 + l_2 c_1 c_2 \\ l_1 s_1 + l_2 s_1 c_2 \\ l_2 s_2 \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

$$\frac{d}{dt} {}^0_3\vec{O} = \frac{d}{d\theta_1} {}^0_3\vec{O} \dot{\theta}_1 + \frac{d}{d\theta_2} {}^0_3\vec{O} \dot{\theta}_2 + \frac{d}{d\theta_3} {}^0_3\vec{O} \dot{\theta}_3$$

$$= {}^0J(\theta) \dot{q} \quad \text{其中 } q = \begin{bmatrix} \theta_1 \\ \theta_2 \\ \theta_3 \end{bmatrix}$$

$$\text{则 } {}^0J(\theta) = \begin{bmatrix} -l_1 s_1 & -l_2 s_2 c_2 & -l_2 s_2 s_2 \\ l_1 c_1 + l_2 c_1 c_2 & -l_2 s_1 s_2 & 0 \\ 0 & l_2 c_2 & 0 \end{bmatrix}$$

$${}^0_3\vec{V} = \begin{bmatrix} V_x \\ V_y \\ V_z \end{bmatrix} = {}^0J(\theta) \dot{q} = \begin{bmatrix} (-l_1 s_1 - l_2 s_2 c_2) \dot{\theta}_1 - l_2 c_1 s_2 \dot{\theta}_2 \\ (l_1 c_1 + l_2 c_1 c_2) \dot{\theta}_1 - l_2 s_1 s_2 \dot{\theta}_2 \\ l_2 c_2 \dot{\theta}_2 \end{bmatrix}$$

5.19 解:

$$\frac{d}{dt} {}^0P_{2ORG} = {}^0J(\theta) \dot{q}$$

$${}^0J(\theta) = \begin{bmatrix} -a_1 s_1 - d_2 c_1 & 0 \\ a_1 c_1 - d_2 s_1 & 0 \\ 0 & 0 \end{bmatrix}$$

$${}^0J(\theta) = \begin{bmatrix} -a_1 s_1 - d_2 c_1 & -s_1 \\ a_1 c_1 - d_2 s_1 & s_1 \\ 0 & 0 \end{bmatrix}$$

由于 ${}^0J(\theta)$ 非方阵, 因此奇异条件为 $\det[{}^0J^T(\theta) {}^0J(\theta)] = 0$

$${}^0J^T(\theta) {}^0J(\theta) = \begin{bmatrix} a_1^2 + d_2^2 & a_1 \\ a_1 & 1 \end{bmatrix}$$

$$\det[{}^0J^T(\theta) {}^0J(\theta)] = a_1^2 + d_2^2 - a_1^2 = d_2^2 = 0$$

即 $d_2 = 0$

当 $d_2 = 0$ 时机器人 ~~处于奇异构型~~ **运动学奇异**

5.29

关节静力矩 $\tau = J^T(\theta) F$

$$= \begin{bmatrix} -l_1 s_1 - l_2 s_{12} & l_1 c_1 + l_2 c_{12} \\ -l_2 s_{12} & l_2 c_{12} \end{bmatrix} \begin{bmatrix} 5 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} -5(l_1 s_1 + l_2 s_{12}) + 3(l_1 c_1 + l_2 c_{12}) \\ -5l_2 s_{12} + 3l_2 c_{12} \end{bmatrix} = \begin{bmatrix} (-5s_1 + 3c_1)l_1 + (-5s_{12} + 3c_{12})l_2 \\ (-5s_{12} + 3c_{12})l_2 \end{bmatrix}$$